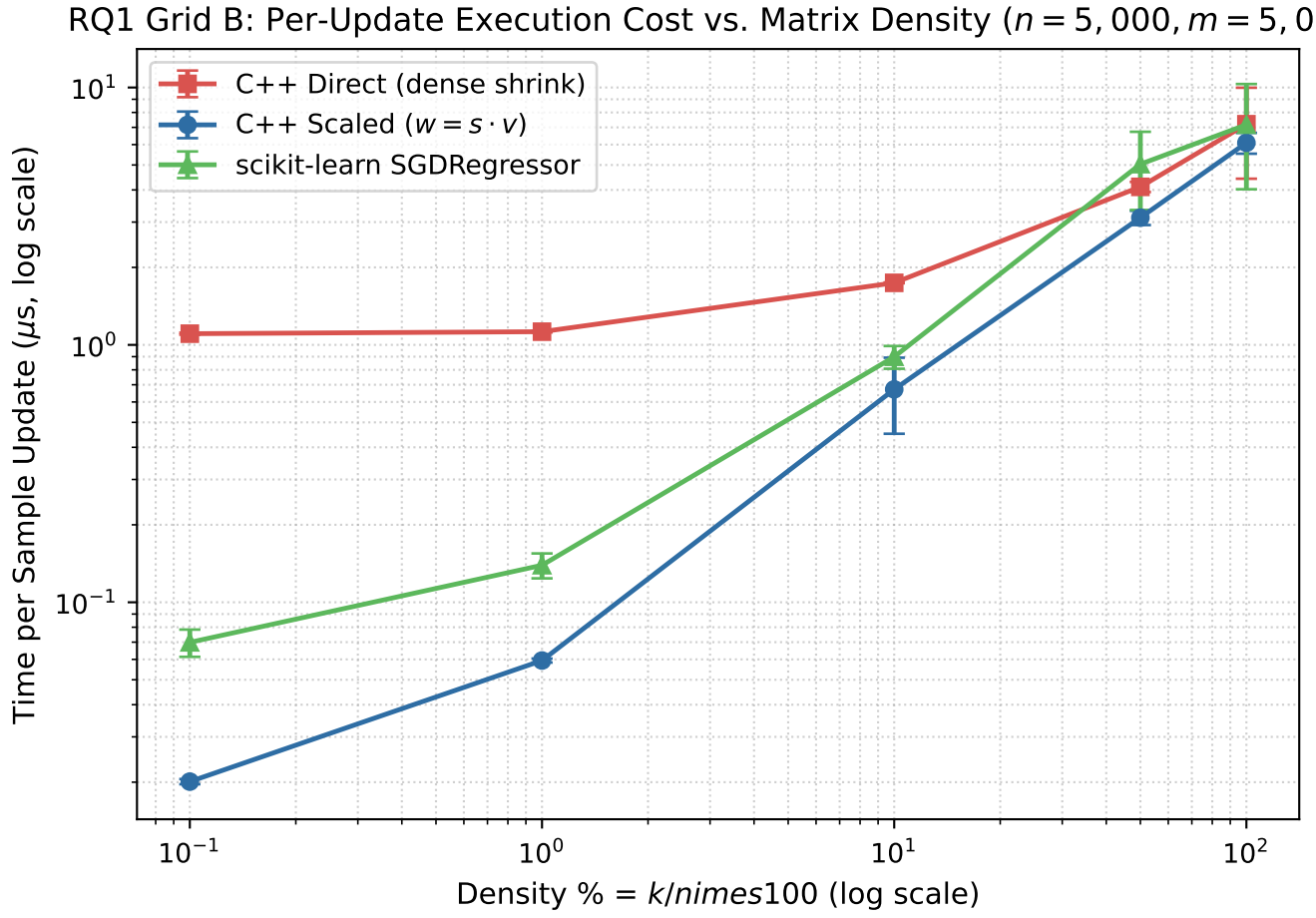
RQ1 Grid B: Per-Update Execution Cost vs. Matrix Density ($n = 5,000, m = 5,00$
C++ Direct (dense shrink)
C++ Scaled ($w = s \cdot v$)
scikit-learn SGDRegressor
Time per Sample Update (μs, log scale)
10^{1}
10^{0}
10^{-1}
10^{-1}
10^{0}
10^{1}
10^{2}
Density % = $k/nimes100$ (log scale)